

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

- To solve (a), we need to find the heat required to increase the temperature of 1 0 0 0 grams of water from 3 0 Â°C to 6 0 Â°C .

- The formula to find the heat is $Q = \text{mass} \times \text{specific heat} \times \Delta T$

- For water , $Q = 1\,000\text{ g} \times 1\text{ cal/g}^\circ\text{C} \times (60^\circ\text{C} - 30^\circ\text{C}) = 1\,000 \times 1 \times 30 = 30\,000\text{ cal} = 30\text{ Kcal}$

- For (b), we need to find the heat required to heat 1 0 0 0 grams (1 kg) of aluminum from 3 0 Â°C to 6 0 Â°C .

- Using the same formula , $Q = 1\,000\text{ g} \times 0.215\text{ cal/g}^\circ\text{C} \times (60^\circ\text{C} - 30^\circ\text{C}) = 1\,000 \times 0.215 \times 30 = 6\,450\text{ cal} \approx 6.45\text{ Kcal}$

- Both values are taken to the nearest Kcal for the options provided .

</think>

<answer> 30 Kcal , 6 . 45 Kcal </answer>